Supplementary Material: Map Growth and Policy Routing in Budgeted Online RGB-D 3DGS

Anonymous ICCV submission

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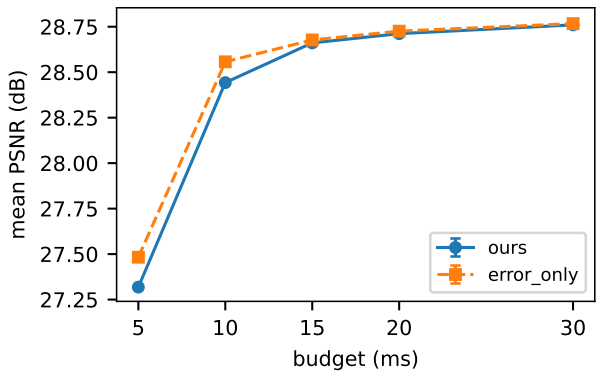


Figure S1. **Budget curve** (fr2_xyz, 30 frames×3, matched modeled budgets): policies are similar across the tested budget range at this short horizon. This experiment does not establish where the 150-frame policies separate as a function of budget.

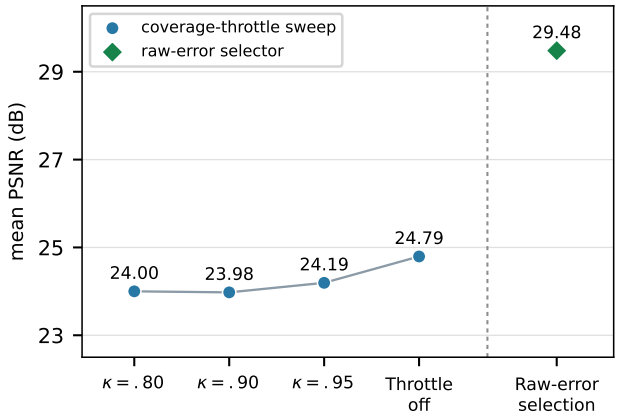


Figure S2. **Replica office0 ablation** (30 frames, $n = 3$). Disabling coverage throttling changes mean PSNR by 0.82 dB; switching from temporal×influence to raw-error changes it by a further 4.69 dB. These are descriptive short-run differences without a reported significance test.

A. Short-horizon ablations

B. Historical and Cross-System Context

The following tables use different substrates or protocols from the corrected matched-budget benchmark. **Historical component ablation (Tab. S1, pose-bugged substrate, 150 frames, 5 seeds).** Relative to A0, A2 (coverage) differs by +1.04 dB and A3 (coverage plus backlog) by +1.08 dB. The direct A3–A2 increment is +0.04 dB; no direct significance test is reported, and comparisons against A0 have Holm-adjusted $p = 0.25$. These numbers are not transferred to the corrected benchmark. A native local SplaTAM run on the same sequence (144f, seed 42: final 27.20 / mean 23.03 dB, SSIM 0.974, LPIPS 0.046, depth 1.26 cm, ATE 0.41 cm) uses a different resolution and tracking setup; these figures are contextual only and do not test compute dilution.

Condition	Mean	Δ vs A0	Map	Finding
A0 baseline	16.07	—	140,998	bloated
A1 + warmup	15.32	−0.75	140,038	harmful
A2 + coverage	17.11	+1.04	44,865	ns vs A0
A3 + backlog	17.15	+1.08	43,824	+0.04 vs A2
A4 + learned	14.53	−1.54	38,317	harmful

Table S1. **Historical component ablation on the pose-bugged substrate** (5 seeds). Values and tests are not evidence for effects on the corrected benchmark. `results/step1_ablation_5seeds/`.

C. Additional Evaluation

D. Further Experiments

The following experiments would refine the mechanism and extend its scope. **Map-growth mediation:** vary continuous spawn rates and pruning strength while holding the candidate error threshold fixed; compare trajectories at matched

Method	Replica	fr2_xyz	Protocol
MonoGS	37.50	16.17	full seq + refine
Point-SLAM	35.17	—	full seq + refine
RTG-SLAM	32.79	17.08	full seq + refine
SplaTAM	—	24.50	full seq + refine
SplaTAM (local run, 144f)	—	27.20 final / 23.03 mean	SLAM 640×480, no refine
Photo-SLAM	—	21.07	full seq + refine
Ours (error_only)	20–33	24.07	budgeted, 100–150f
Ours (throttled)	20–29	**27.64**	budgeted, 100–150f

Table S2. **Published-SOTA context (protocol-mismatched) and one local SplaTAM run.** Published results are shown for context, not as matched comparisons. The local run uses the same sequence but a different resolution and tracking setup; details are in the text.

Method	fr2_xyz	fr1_xyz	Horizon	System?
Ours (GN VO)	0.9	2.0	100f	no (baseline)
MonoGS	1.71	—	full	yes
SplaTAM	0.50	—	full	yes
Photo-SLAM	0.63	—	full	yes

Table S3. **ATE RMSE (cm).** Rows 2–4 are published full-system numbers; ours is a decoupled odometry sanity check on fixed horizons.

map size and depth quality. A budget proportional to current map size would also test whether the fourfold rescue persists under a less coarse intervention. **Router generalization:** obtain a third sensor domain and scenes outside the reported set for threshold calibration; then test structured missing-depth artifacts and geometry-preserving TUM hole repair. The present random-hole and valid-depth-smoothing interventions did not reverse the preferred policy. **Longer mapping evaluation:** use sequences of at least 500 frames, multiple start windows, a real tracker, and depth or geometric metrics. **Corrected utility labels:** regenerate counterfactual labels on the corrected pose substrate before drawing conclusions about learned ranking. **Finer-grained adaptivity:** the 7-scene routing table shows motion/exploration regime matters within a sensor (fr1_room, fr1_rpy trail by $\approx -1.1\,\mathrm{dB}$); extending the observable cue beyond hole-rate (e.g. optical-flow magnitude, coverage velocity) could route rapid-motion frames to instantaneous error while keeping temporal smoothing elsewhere.